Steering Wheel & Display

Set Text Position

Send the sequence to steering wheel

Command byte	x upper	x lower	y upper	y lower	Parity

Command byte	Bit 7: 0b1 Bit 6-2: text slot (0-31 inclusive) Bit 1-0: 0b00
x	x position on the screen. $0 \le x \le 536$ Split between 2 bytes
y	y position on the screen, aligned to memory bank on display hardware so the range is smaller. $0 \le y \le 42$ Split between 2 bytes
Parity	Bitwise XOR of all previous bytes

Set Text Size

Send the sequence to steering wheel

Command byte	Size	Parity

Command byte	Bit 7: 0b1 Bit 6-2: text slot (0-31 inclusive) Bit 1-0: 0b01
Size	Text size. $0 \le size \le 15$

Parity	Bitwise XOR of all previous bytes

Set Text Mode (Flashing)

Send the sequence to steering wheel

Command byte	Mode	Parity

Command byte	Bit 7: 0b1 Bit 6-2: text slot (0-31 inclusive) Bit 1-0: 0b10
Mode	Text flash frequency. 0 ≤ mode ≤ 7 0: not flashing 1: highest frequency 7: lowest frequency
Parity	Bitwise XOR of all previous bytes

Draw Word

Send the sequence to steering wheel

Command byte	Length	Characters[]	Parity

Command byte	Bit 7: 0b1 Bit 6-2: text slot (0-31 inclusive) Bit 1-0: 0b11
Length	Number of characters
Characters[]	Send each character as an ASCII byte
Parity	Bitwise XOR of all previous bytes

Set Display Brightness

Send the sequence to steering wheel

Command byte	Brightness	Parity

Command byte	0x03
Brightness	Display brightness. 0 ≤ brightness ≤ 100
Parity	Bitwise XOR of all previous bytes

Set Indicator Light

Send the sequence to steering wheel

Command byte	Indicator lights	Parity

Command byte	0x02
Indicator lights	Bit 7-2: Unused Bit 1: Left indicator light Bit 0: Right indicator light
Parity	Bitwise XOR of all previous bytes

Set Fault Light

Send the sequence to steering wheel

Command byte	Fault light	Parity

Command byte	0x01
Fault light	Bit 7-1: Unused Bit 0: Fault light

Parity	Bitwise XOR of all previous bytes

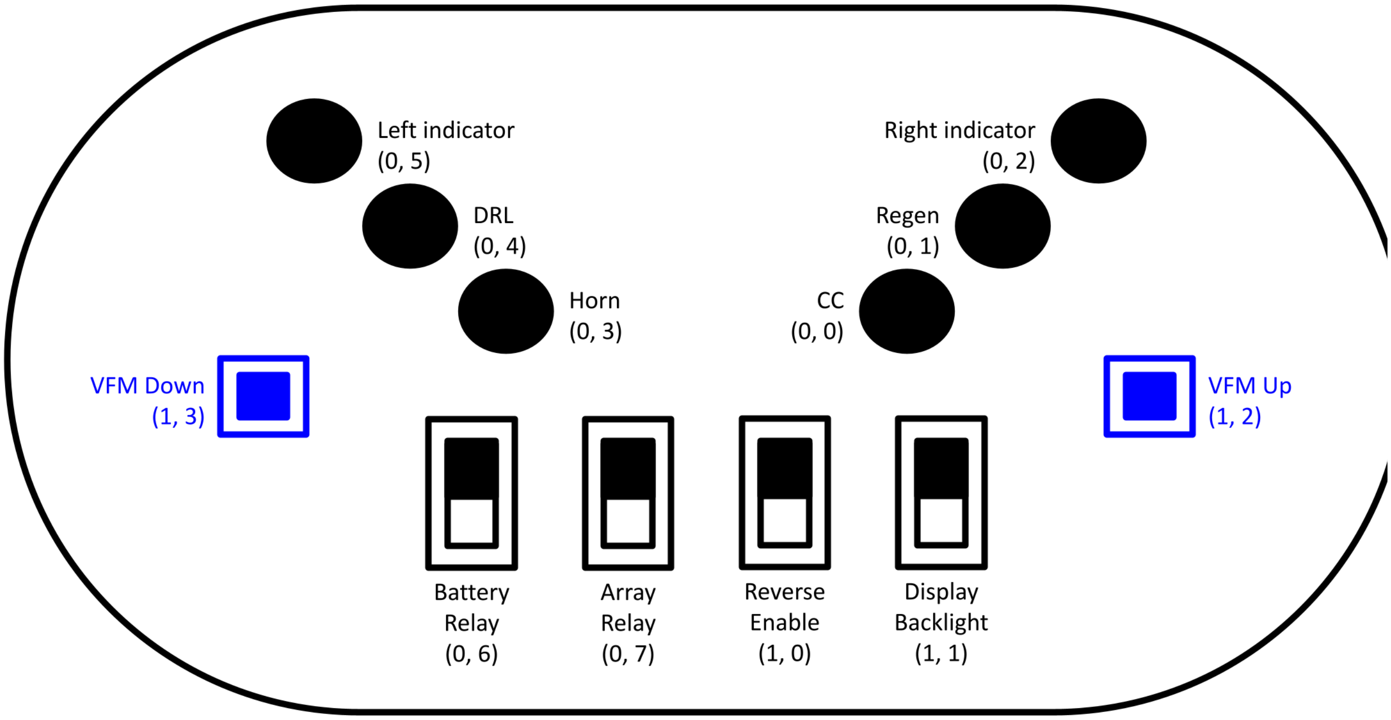
Read Input

Send 5 dummy bytes, and gets 5 bytes back from steering wheel

Start marker	Data byte 0	Data byte 1	Parity	End marker

Start marker	0b10101010 Used to determine shift in byte order and reorder byte sequence
Data byte 0 (active low)	Bit 7: Array relay Bit 6: Battery relay Bit 5: Left indicator Bit 4: DRL (daytime running light) Bit 3: Horn Bit 2: Right indicator Bit 1: Regen (regenerative breaking) Bit 0: CC (cruise control)
Data byte 1 (active low)	Bit 7-4: Unused Bit 3: VFM (variable field magnet) down Bit 2: VFM up Bit 1: Reverse enable Bit 0: Display backlight
Parity	Data byte 0 ^ Data byte 1 (bitwise XOR)
End marker	0b01010101 Used to determine shift in byte order and reorder byte sequence

Note that for push buttons there is no latch, so the bit value only shows whether the button is pushed at the moment. Toggling logic needs to be handled elsewhere.



(Byte, Bit)
Blue: back side push button

+ ALT